



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions.

Dataset at a Glance

3,900

Purchases

18

Features

\$59.76

Avg. Spend

3.75

Avg. Rating

Demographics

Age, Gender, Location, Subscription

Purchase Details

Category, Amount, Season, Size, Color

Behavior

Discounts, Frequency, Ratings, Shipping

Data Preparation in Python



01

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

Clean & Impute

Filled 37 missing Review Ratings with category median

03

Feature Engineering

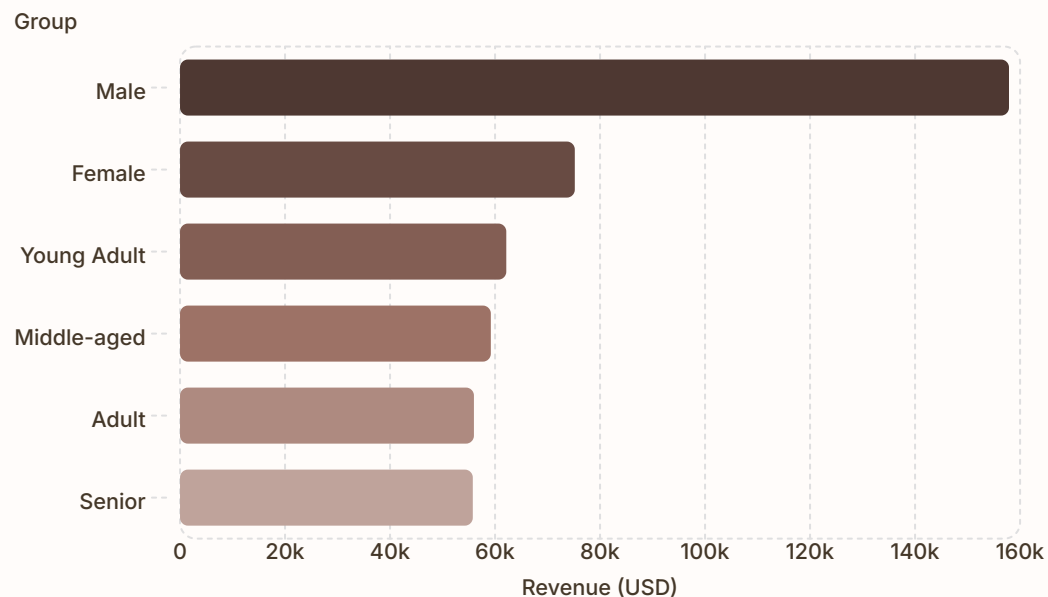
Created `age_group` and `purchase_frequency_days` columns

04

Load to PostgreSQL

Cleaned DataFrame exported to database for SQL analysis

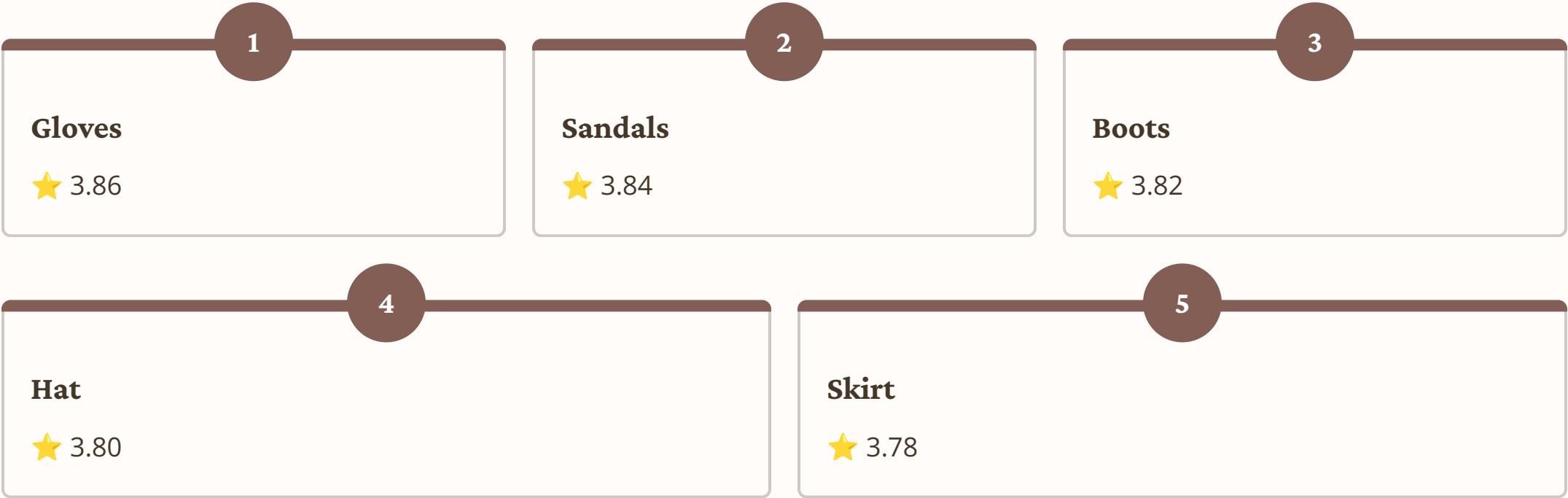
Revenue & Spending Insights



Key Findings

- Male customers generate **2× more revenue** than female
- Young Adults are the **top revenue age group** at \$62,143
- Express shipping users spend slightly more: **\$60.48 vs. \$58.46** (Standard)
- 839 discount users still spent **above average**

Top-Rated Products



Best-Selling Products by Category

ACCESSORIES



Jewelry (171)
Sunglasses (161)
Belt (161)

CLOTHING



Blouse (171)
Pants (171)
Shirt (169)

FOOTWEAR



Sandals (160)
Shoes (150)
Sneakers (145)

OUTERWEAR



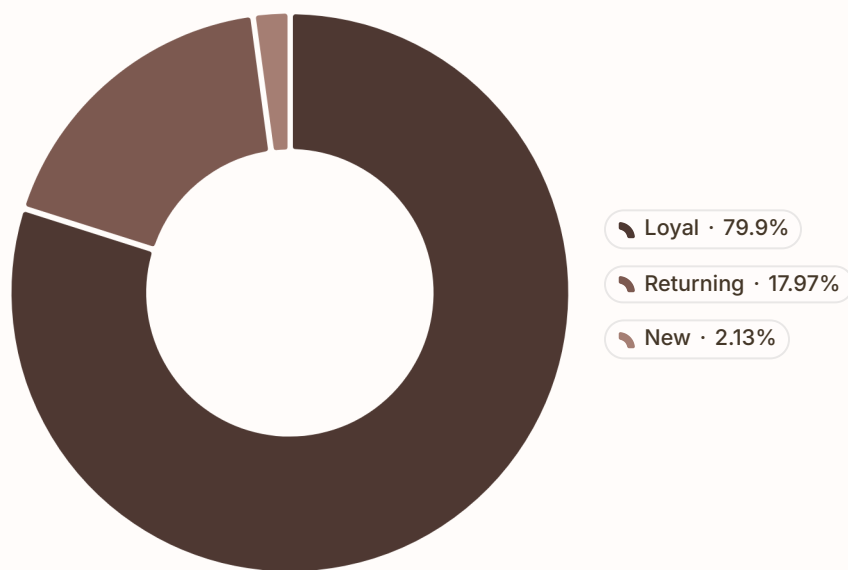
Jacket (163)
Coat (161)

Category Leaders

- **Accessories:** Jewelry tops at 171 orders
- **Clothing:** Blouse & Pants tied at 171
- **Footwear:** Sandals lead with 160 orders
- **Outerwear:** Jacket dominates at 163 orders

i Hats have the highest discount rate at **50%**, followed by Sneakers (49.66%) and Coats (49.07%)

Customer Segmentation



Segment Breakdown

- **Loyal:** 3,116 customers — the vast majority
- **Returning:** 701 customers
- **New:** only 83 customers

Subscription Snapshot

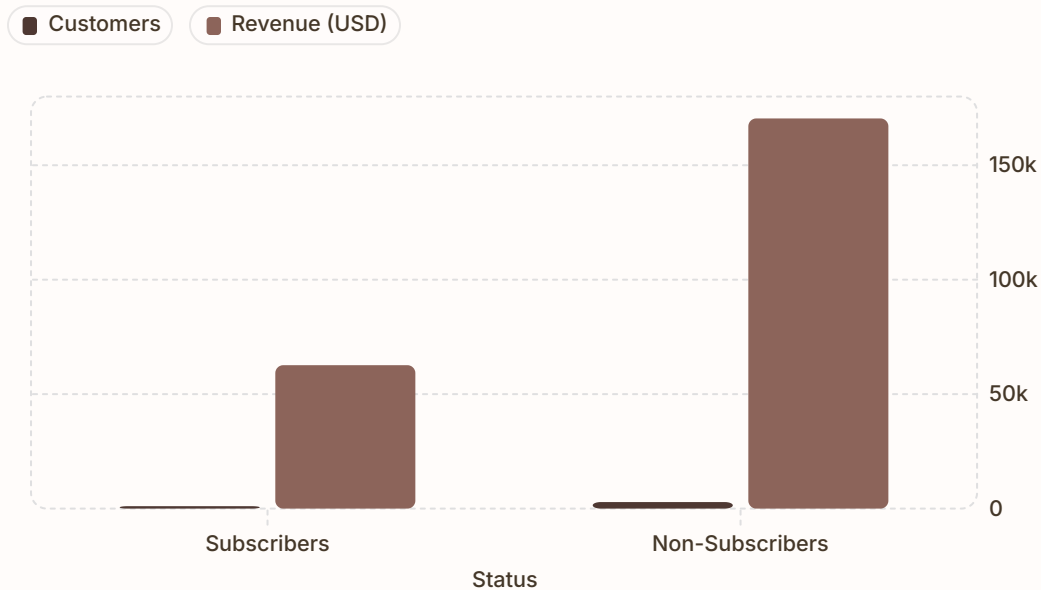
- Subscribers: 1,053 — avg spend **\$59.49**
- Non-subscribers: 2,847 — avg spend **\$59.87**
- Repeat buyers (>5 purchases): **958 subscribed** vs. 2,518 not

Power BI Dashboard



3.9K Customers	\$59.76 Avg. Purchase
3.75 Avg. Rating	27% / 73% Subscribers / Non

Subscription & Discount Dynamics



Discount-Heavy Products

- Hat — **50%** discounted purchases
- Sneakers — **49.66%**
- Coat — **49.07%**
- Sweater — **48.17%**
- Pants — **47.37%**



High discount dependency may erode margins — review pricing strategy

Business Recommendations



Boost Subscriptions

Promote exclusive benefits to grow the 27% subscriber base



Loyalty Programs

Reward repeat buyers to accelerate movement into the Loyal segment



Review Discount Policy

Balance sales boosts with margin control on high-discount products



Targeted Marketing

Focus on Young Adults and express-shipping users — highest revenue potential

